

Bastian Kuth

Dr.-Ing. (PhD)

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Work Experience

- 05/2026 – today **Senior Software Development Engineer** *full-time, remote*
Advanced Micro Devices (AMD)
Supervisor: Dominik Baumeister 
- 01/2023 – 03/2026 **Research Associate** *full-time*
Coburg University of Applied Sciences
Supervisor: Prof. Dr.-Ing. Quirin Meyer  
Project GeoFlow – Data-centered real-time geometry processing on graphics cards
Worked closely with AMD team to research novel techniques for real-time geometry processing using Direct3D12 GPU work graphs and mesh shaders.
Delivered prototypes, benchmarks, tutorials, demos (GDC), courses (GPC, SIGGRAPH), articles (GPUOpen), and four award-winning papers (HPG, VMV, C&G) featuring these technologies.
- 03/2024 – 08/2024 **Contract Lecturer** 2 SWS
Coburg University of Applied Sciences
3D Gaussian Splatting in Direct3D12
Supervised visual computing students developing a Gaussian splatting renderer in Direct3D12 and gave lectures about real-time rendering.
- 03/2022 – 12/2022 **Research Assistant** *part-time*
Nuremberg Institute of Technology: [Game Tech Lab](#)
Supervisor: Prof. Dr. rer. nat. Bartosz von Rymon Lipiński  
Project IBMPP – Image-Based Media Production Pipeline
Conceptualized and implemented a VR renderer and genetic optimization pipeline for densely feature-matched images (OpenGL 4.5, OpenXR, C++). Co-authored two papers.
- 02/2020 **Research Assistant** *part-time*
Coburg University: [mixedrealitylab](#)
Supervisor: Prof. Dr. techn. Jens Grubert  
Project Virtual Office
Developed VR prototypes (Unity, C#). Assisted in VR user studies. Co-authored a paper.
- 10/2019 – 02/2020 **Internship Semester** *full-time*
Siemens Healthineers – Product Development
Supervisor: Dr.-Ing. Philip Mewes 
Project [AI-Rad Companion](#)
Responsible for technical setups of user studies required for FDA approval of medical products. Setup included a simulated hospital environment with PACS, [teamplay](#), and AI Rad Companion platform to be used by Radiologists in these studies.
- 01/2019 – 03/2019 **Research Assistant** *part-time*
Coburg University: [mixedrealitylab](#)
Supervisor: Prof. Dr. techn. Jens Grubert  
Project Virtual Office
Developed VR prototypes (Unity, C#), installed an OptiTrack MoCap system, co-authored a paper.
- 03/2018 – 09/2018 **Working Student**
- 05/2017 – 09/2017 **Summer Intern** 
Siemens Healthineers – Product Innovation
Supervisor: Dr.-Ing. Peter Fischer 
Project Surgery Robotics
Evaluated the feasibility of 3D-printed X-ray markers for surgical robotics (C++, Python, OpenCV, ViSP). Generated synthetic X-ray data (Blender). Was entrusted with independent operation of an ARTIS pheno X-ray system.

Education

- 2023 – Present **Doktor der Ingenieurwissenschaften (PhD)**
University of Erlangen–Nuremberg (FAU) • Coburg University of Applied Sciences
Thesis: Real-Time Geometry Amplification on Graphics Hardware
Advisors: Prof. Dr.-Ing. Marc Stamminger  • Prof. Dr.-Ing. Quirin Meyer 
Grade: with distinction (best possible grade)
- 2021 – 2022 **Master of Science in Computer Science**
Nuremberg Institute of Technology Georg Simon Ohm
Thesis: Real-Time Image-Based 3D-VR-Rendering of Open Environments
Advisor: Prof. Dr. rer. nat. Bartosz von Rymon Lipiński 
Thesis Grade: 1.0 Overall Grade: 1.0 (best of the year)
- 2017 – 2021 **Bachelor of Science in Computer Science**
Coburg University of Applied Sciences
Thesis: Compression of Vertex Blending Attributes
Advisor: Prof. Dr.-Ing. Quirin Meyer 
Thesis Grade: 1.0 Overall Grade: 1.3

Skills

Programming

Proficient in: C++, C, Python, HLSL, GLSL • Familiar with: Java, C#, Haskell, JavaScript

Tools, Libraries and Software

Blender, Git, Gimp, L^AT_EX, Overleaf, Gurobi, SCIPopt, Direct3D12, OpenGL, OpenCV, OpenXR, Vulkan, CUDA, CMake, Unity, NumPy, SymPy, ImGui, Real Virtuality Engine, Matplotlib, Tensorflow, Keras, Pytorch, Adobe Premiere, Visual Studio, VS Code

Languages

German (native), English (fluent)

Awards

- 11/2025 **Best Paper Award (2nd place)**
At [Journées françaises de l'informatique graphique 2025](#)
- 06/2025 **Wolfgang Straßer Best Paper Award (2nd place)**
At [High-Performance Graphics 2025](#)
For *Real-Time GPU Tree Generation*
- 09/2024 **Best Paper Award**
At [Vision, Modeling, and Visualization 2024](#)
For *Towards Practical Meshlet Compression*
- 07/2024 **Wolfgang Straßer Best Paper Award (1st place)**
At [High-Performance Graphics 2024](#)
For *Real-Time Procedural Generation with GPU Work Graphs*
- 06/2023 **Best Computer Science Master's Degree of the Year**
At [Nuremberg Institute of Technology Georg Simon Ohm](#)
- 06/2023 **Wolfgang Straßer Best Paper Award (3rd place)**
At [High-Performance Graphics 2023](#)
For *Edge-Friend: Fast and Deterministic Catmull-Clark Subdivision Surfaces*
- 06/2023 **DATEV Sponsorship Award**
At [Nuremberg Institute of Technology Georg Simon Ohm](#)
For Master's Thesis *Real-Time Image-Based 3D-VR-Rendering of Open Environments*
- 05/2022 **Best Paper Award**
At [ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games 2022](#)
For *Permutation Coding for Vertex-Blend Attribute Compression*
- 07/2021 **Wolfgang Straßer Best Paper Award (1st place)**
At [High-Performance Graphics 2021](#)
For *Vertex-Blend Attribute Compression*
- 10/2019 **Best Paper Award**
At [International Symposium on Mixed and Augmented Reality 2019](#) (IEEE ISMAR)
For *ReconVigURation: Reconfiguring Physical Keyboards in Virtual Reality*

Voluntary Work

- 2024 – Present **Paper Reviewer** at JCGT (1), HPG 2026 (3), Eurographics 2026 (2), HPG 2025 (3), Eurographics 2025 (1), SIGGRAPH Asia 2024 (1), HPG 2024 (2)
- 09/2016 – 03/2017 **Bundesfreiwilligendienst** *full-time*
Federal Agency for Technical Relief (THW) • Geschäftsstelle Bamberg

Video Game Modding

- 2019–2021 **Nassau 1715**
Our international team created a total conversion mod that transforms Arma 3 from a modern military simulation on a Greek island into one about pirates in the Bahamas. I handled most of the programming. Key features include a physically accurate sailing model and a custom interaction system. The mod has ~62k total downloads on the [Steam Workshop](#).
- 2017 **WMO – Walkable Moving Objects**
A mod that extends Arma 3's Real Virtuality Engine to support players riding moving objects, such as cars, boats, or airplanes. With the vanilla game engine, players just slide off or die when trying to board a vehicle. This mod fixes this. It currently has ~263k+ subscribers and ~538k+ total downloads on the [Steam Workshop](#).
- 2014–2019 **Loewenherz Altis Life**
Volunteered as developer, admin, and moderator for one of Germany's largest Altis Life servers – a 100+ player (MMO)RPG game mode for Arma 3. Contributed to scripting, texturing, 3D modeling, as well as gameplay/sound/level design. At its peak, the server database had 70,000 unique players and the community forum over 7,000 members.

Notable Media Coverage

- 2025 **TechPowerUp** [link](#)
Researchers Unveil Real-Time GPU-Only Pipeline for Fully Procedural Trees
- 2025 **PC Games Hardware** [link](#)
Von 35 GiB auf 51 KiB: Forscher demonstriert "Work Graphs" für 3D-Rendering von Bäumen
- 2025 **PC Gamer** [link](#)
Graphics researchers have created a GPU-run procedural algorithm for creating an equivalent 35.6 GB worth of trees, leaves, and bushes from just 52 kB of data
- 2025 **Tom's Hardware** [link](#)
AMD researchers reduce graphics card VRAM capacity of 3D-rendered trees from 38GB to just 52 KB with work graphs and mesh nodes – shifting CPU work to the GPU yields tremendous results
- 2024 **Heise Online** [link](#)
New GPU technology creates dynamic 3D worlds
- 2024 **Tom's Hardware** [link](#)
AMD shows off DX12-related rendering advances that make game engines more efficient and less dependent on the CPU – demo shows a 64% improvement

Public Talks

- 2025 **GPU Work Graphs**
Course at SIGGRAPH 2025 in Vancouver, Canada, [recording](#)
- 2025 **Real-Time GPU Tree Generation**
Paper presentation at HPG 2025 in Copenhagen, Denmark, [recording](#)
- 2024 **GPU Work Graphs - mastering the future of GPU programming**
Masterclass at GPC 2024 in Breda, Netherlands
- 2024 **Towards Practical Meshlet Compression**
Paper presentation at VMV 2024 in Garching, Germany
- 2024 **Real-Time Procedural Generation with GPU Work Graphs**
Paper presentation at HPG 2024 in Denver, USA, [recording](#)
- 2023 **Edge-Friend: Fast and Deterministic Catmull-Clark Subdivision Surfaces**
Paper presentation at HPG 2023 in Delft, Netherlands, [recording](#)

Publications

- 2026 **GPU View-Adaptive Crack-Free Subdivision of Bézier Surfaces**
Bastian Kuth^{*}, Seyedmasih Tabaei^{*}, Max Oberberger, Quirin Meyer
Eurographics Symposium on Rendering - Symposium Papers (EGSR)
^{*}joint first authors
- 2025 **GPU Work Graphs**
Bastian Kuth, Max Oberberger, Quirin Meyer
SIGGRAPH Conference Courses 2025
- 2025 **Real-Time GPU Tree Generation**
Bastian Kuth, Max Oberberger, Carsten Faber, Pirmin Pfeifer, Seyedmasih Tabaei, Dominik Baumeister, Quirin Meyer
High-Performance Graphics - Symposium Papers (HPG)
- 2025 **EGO-VC – Evolutionary GPU-Optimization of Visual Correspondences for Image Alignment**
Thomas Chang, Karl Hartmann, Bastian Kuth, Simon Seibt, Bartosz von Rymon Lipiński
WSCG 2025 (follows section 3 of my master thesis)
- 2025 **Multidimensional Image Morphing-Fast Image-based Rendering of Open 3D and VR Environments**
Simon Seibt, Bastian Kuth, Bartosz von Rymon Lipiński, Thomas Chang, Marc Erich Latoschik
Virtual Reality and Intelligent Hardware (VRIH) (follows section 2 of my master thesis)
- 2025 **Real-Time Meshlet Decompression**
Bastian Kuth, Max Oberberger, Felix Kawala, Sander Reitter, Sebastian Michel, Matthäus Chajdas, Quirin Meyer
Computers & Graphics
- 2024 **Towards Practical Meshlet Compression**
Bastian Kuth, Max Oberberger, Felix Kawala, Sander Reitter, Sebastian Michel, Matthäus Chajdas, Quirin Meyer
Vision, Modeling, and Visualization (VMV)
- 2024 **Real-Time Procedural Generation with GPU Work Graphs**
Bastian Kuth, Max Oberberger, Carsten Faber, Dominik Baumeister, Matthäus Chajdas, Quirin Meyer
Proceedings of ACM on Computer Graphics and Interactive Techniques (PACMCGIT / HPG)
- 2023 – 2024 **Mesh shaders on RDNA™ graphics cards**
Max Oberberger, Bastian Kuth, Quirin Meyer
GPUopen (not peer-reviewed)
- 2023 **Edge-Friend: Fast and Deterministic Catmull-Clark Subdivision Surfaces**
Bastian Kuth, Max Oberberger, Matthäus Chajdas, Quirin Meyer
Computer Graphics forum (CGF / HPG)
- 2022 **Permutation Coding for Vertex-Blend Attribute Compression**
Christoph Peters, Bastian Kuth, Quirin Meyer
ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)
- 2021 **Vertex-Blend Attribute Compression**
Bastian Kuth, Quirin Meyer
High-Performance Graphics - Symposium Papers (HPG)
- 2020 **Breaking the Screen: Interaction Across Touchscreen Boundaries in Virtual Reality for Mobile Knowledge Workers**
Verena Biener, Daniel Schneider, Travis Gesslein, Alexander Otte, Bastian Kuth, Per Ola Kristensson, Eyal Ofek, Michel Pahud, Jens Grubert
IEEE Transactions of Visualization and Computer Graphics (TVCG)
- 2019 **ReconVigURation: Reconfiguring Physical Keyboards in Virtual Reality**
Daniel Schneider, Alexander Otte, Travis Gesslein, Philipp Gagel, Bastian Kuth, Mohamad Shahm Damlakhi, Oliver Dietz, Eyal Ofek, Michel Pahud, Per Ola Kristensson, Jörg Müller, Jens Grubert
IEEE Transactions of Visualization and Computer Graphics (TVCG)